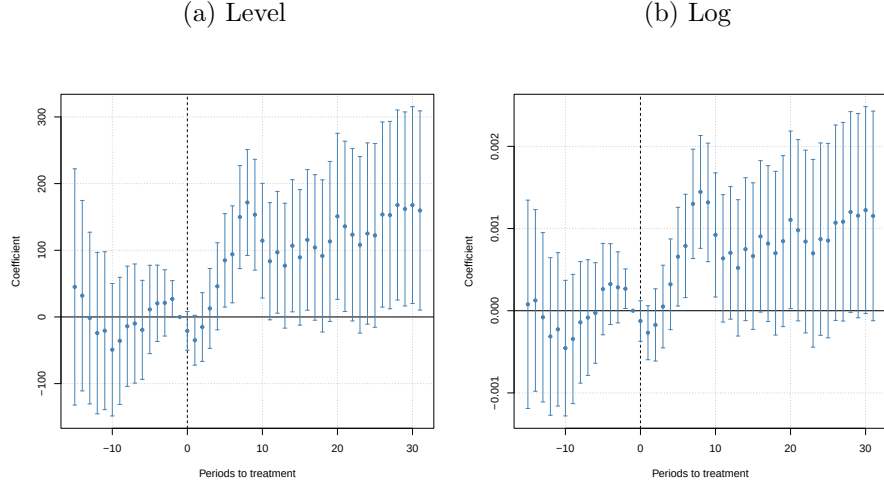


Figure 3: Effect of LLM exposure on firm-level compensation



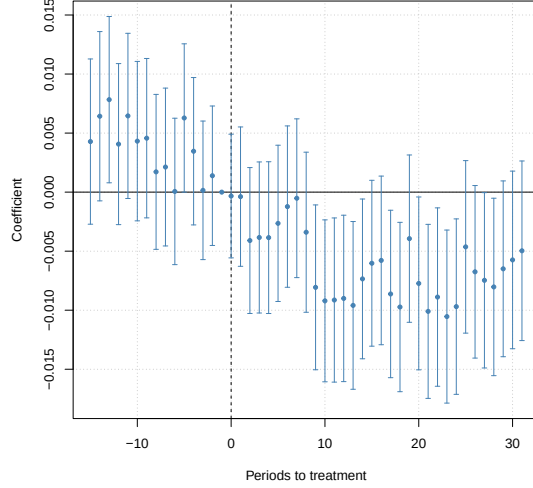
Notes: This figure plots event study estimates of the effect of LLM exposure on average compensation within UK firms. Compensation is defined as annual base salary plus bonuses. Panel A shows compensation in levels, Panel B in logs. Coefficients are estimated from Equation (3) and are shown relative to October 2022, the final pre-treatment month. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

Figure 4 presents event study estimates for the probability that exposed vs. less exposed firms post any job vacancy in a given month. The results show a change in hiring patterns following November 2022. Prior to this date, firms with different exposure levels did not have statistically significantly different trends in job postings.⁸ Following the release of ChatGPT, however, we observe that a one standard deviation increase in LLM exposure reduces firms' likelihood of posting new jobs by 1.0 percentage point ($p < 0.001$, see Table A1). This effect emerges gradually and becomes statistically significant after approximately 10 months, mirroring the timeline of employment stock adjustments.

Next, we decompose the aggregate hiring reduction by estimating occupation-specific effects across job categories. We estimate Equation (4) separately for 50 distinct occupational categories, with the outcome variable indicating whether a

⁸Although all pre-trend estimates are statistically insignificant, the point estimates exhibit a slight downward drift. If this reflects a differential pre-trend rather than sampling variation, the post-treatment estimates may overstate the true effect.

Figure 4: Effect of LLM exposure on firm-level job listings



Notes: This figure plots event study estimates of the effect of LLM exposure on the probability that a UK firm posts any job vacancy in a given month. Coefficients are estimated from Equation (3) with firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The dependent variable is an indicator equal to one if the firm posted at least one vacancy in month t . The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

firm posts any vacancy for that specific role in a given month. We continue to compare exposed vs. less exposed firms in the same time period.

Figure 5 presents these occupation-specific treatment effects, ordered by magnitude. The results reveal substantial heterogeneity in hiring adjustments across roles. Technical positions experience the sharpest declines: a one standard deviation increase in firm-level LLM exposure reduces the likelihood of posting software engineer positions by 1.1 percentage points ($p < 0.001$), QA tester positions by 1.0 percentage points ($p < 0.001$), and data analyst positions by 0.8 percentage points ($p < 0.001$). We find similar declines for other technical roles such as IT specialists, IT project managers, and infrastructure engineers. Creative and design roles also see significant reductions, with job postings for designer positions decreasing by 0.6 percentage points ($p < 0.001$) and for producers by 0.5 percentage points ($p < 0.001$). In contrast, sales representative positions show a statistically significant increase of 0.3 percentage points ($p < 0.001$). This pattern suggests that exposed firms are scaling back recruitment for technical and analytical roles where LLMs can augment or